

EC2 Setup: I just used the first EC2 instance we set up together in class. I connected to the instance via SSH in my terminal and launched a Jupyter notebook from it.

Dataset Month: I used January 2026 → Ex. taxi_data/yellow_tripdata_2026-01.parquet

Data Cleaning Assumptions: I removed trips that had a non-positive traveling distance, which is physically impossible. I also removed trips that had a pickup time that happened after the dropoff time, which is also impossible. Trips with negative fares/total_amounts were removed because those should be impossible as well. Trips that lasted longer than 24 hours were probably errors, so they were removed. Trips with missing data, such as missing timestamps, would be meaningless in our analysis, so they were removed.

Question 1 Answer:

```
+-----+-----+
|taxi_type| count|
+-----+-----+
|   yellow|3518128|
|   green| 38909|
+-----+-----+
```

Question 2 Answer:

```
+-----+-----+
|taxi_type| avg(fare_amount)|
+-----+-----+
|   yellow| 21.08260282173478|
|   green|16.003356806908457|
+-----+-----+
```

Question 3 Answer:

```
+-----+-----+
|taxi_type|avg(trip_distance)|
+-----+-----+
|   yellow| 6.757001624157525|
|   green| 12.98990593435966|
+-----+-----+
```

Question 4 Answer:

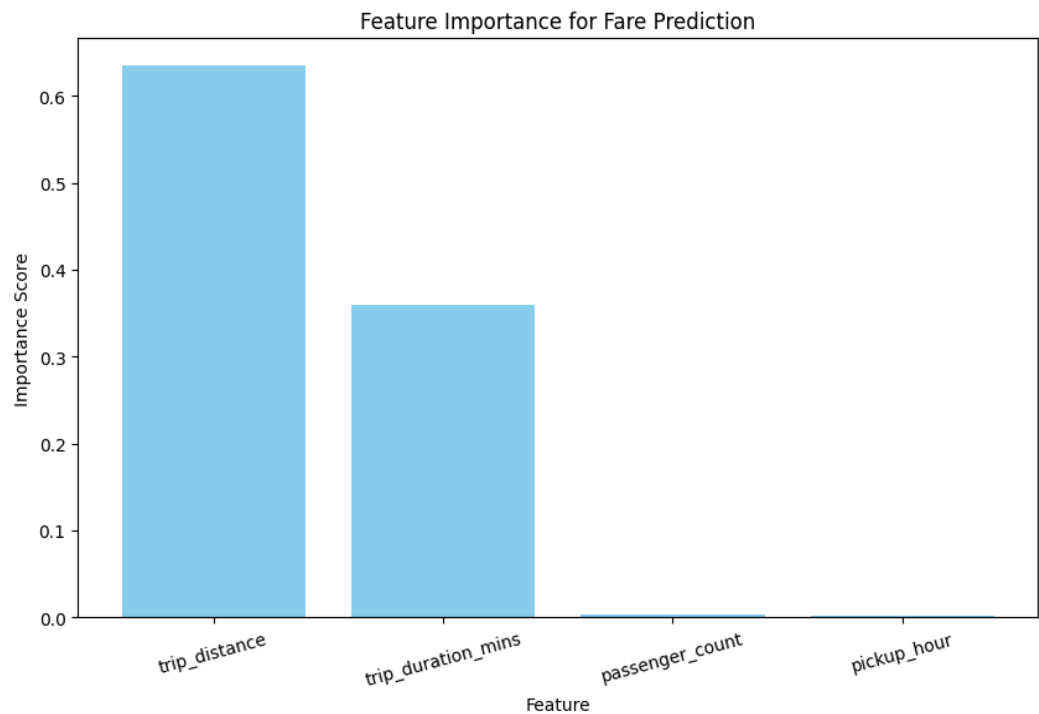
```
+-----+-----+
|pickup_hour| count|
+-----+-----+
|          18|232570|
|          17|224277|
|          15|216305|
|          19|211118|
```

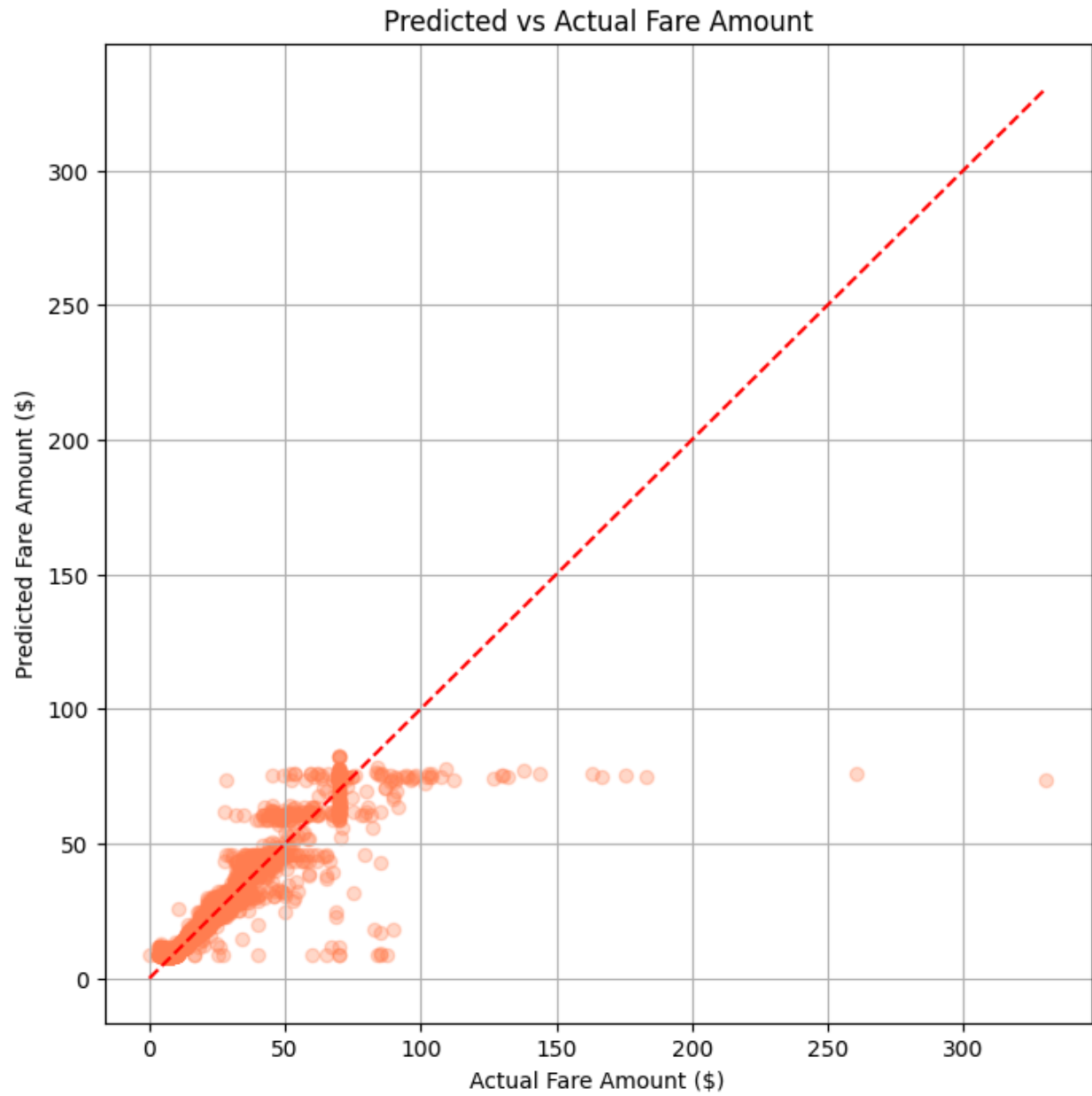
	20		208072	
	21		207428	
	16		206485	
	14		202317	
	13		190970	
	22		190251	
	12		183146	
	11		167923	
	10		158111	
	9		154087	
	23		148822	
	8		146020	
	7		112027	
	0		109013	
	1		75623	
	6		62649	
	2		52470	
	3		37491	
	5		32070	
	4		27792	
+-----+-----+				

Question 8 Answer:

Training RMSE: 7.89

Testing RMSE: 8.28





S3 Bucket URI: <s3://yialex-hw3/nyc-taxi-assignment/>